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Distribution Network Optimization and Flexibility Enhancement Based on Power Grid Equipment Maintenance

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Abstract

With increasing integration of renewable energy, traditional distribution networks face challenges such as low flexibility, poor response speed, and operational inefficiency. To address these issues, this paper proposes a two-layer optimization framework for active distribution networks that integrates grid reconfiguration and equipment maintenance considerations. The upper layer optimizes the network topology and branch flexibility using a flexibility adequacy index and power loss minimization. The lower layer performs distributed robust dispatch under renewable generation uncertainty. A hybrid algorithm combining Ant Colony Optimization (ACO), Fire Hawk Optimization (FHO), and Differential Evolution (DE) is developed to solve the model efficiently. Simulation is conducted on a modified 62-node test system. Comparative results with deterministic, stochastic, and robust models show that the proposed approach achieves the lowest average cost and maximum cost under 500 Monte Carlo scenarios. It also significantly reduces flexibility deficits and renewable curtailment. In addition, the model contributes to predictive maintenance by identifying optimal switching strategies and branch stress levels. These findings demonstrate the method's effectiveness in improving economic efficiency, system flexibility, and equipment sustainability.

Keywords: two-layer optimization framework; distributed robust dispatch; Monte Carlo scenarios; renewable generation uncertainty

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1. Introduction

The rapid increase in renewable energy integration has led to significant structural changes in modern power systems. While promoting clean energy transformation, high renewable energy penetration also brings inherent variability and uncertainty, posing challenges to system stability, operational cost, and supply–demand balance [1,2]. In particular, the distribution network is most impacted due to its proximity to distributed generation sources, such as rooftop photovoltaic (PV) and local wind farms [3]. Many countries are adapting their power system operations and market frameworks to accommodate this shift. For instance, Del Pizzo et al. analyzed the operational adjustments made by the Italian TSO to manage high RES penetration during low-load seasons [4]. Similarly, Wang et al. emphasized the risk of insufficient dispatchable capacity under high renewable conditions, advocating for new capacity pricing strategies [5]. These studies highlight the need for both technical and market mechanisms to ensure adequacy and flexibility in future power systems.

At the distribution level, various studies have explored strategies to handle renewable uncertainty, including scenario-based stochastic programming, robust optimization, and reconfiguration techniques [6–8]. Luo et al. proposed a dynamic reconfiguration strategy that models wind and solar uncertainties through interval prediction and scenario generation [9]. Gong and Wu introduced a risk assessment model combining stochastic power flow and failure probability to quantify operational risk [10]. These works demonstrate the value of modeling uncertainty in renewable-rich distribution networks. Recent studies have also explored distributed optimization frameworks for coordinating user-side resources among multiple stakeholders, such as photovoltaic owners, grid operators, and end-users, aiming to improve source–load matching and enhance grid integration flexibility [11,12]. These works demonstrate the value of modeling uncertainty in renewable-rich distribution networks. Flexibility has emerged as a key metric in the design and operation of modern power systems [13,14]. However, existing flexibility metrics are often static and fail to capture system-wide spatial and temporal constraints [15,16]. Herath et al. proposed a flexibility enhancement process using a Normalized Flexibility Index and Violation Probability metric to guide planning [17]. Huang et al. developed a multi-scale flexibility quantification approach in district heating networks to support EPS scheduling [18].

From a control strategy perspective, robust optimization methods offer strong potential for accommodating uncertainty in real-time operation [19,20]. Yi et al. developed a multi-objective robust dispatch framework incorporating demand response uncertainty [21]. Liao et al. proposed an affine adjustable robust strategy coordinating PV, load, and storage under distribution network uncertainty [22]. Yet, these methods typically treat dispatch decisions independently of network structure, limiting their ability to address topological constraints and flexibility allocation simultaneously. Recently, two-layer optimization frameworks have gained attention due to their ability to separate structural decisions (e.g., network reconfiguration) from operational decisions (e.g., dispatch or control) [23,24]. Huang et al. designed a web-of-cells two-layer model incorporating supply–demand flexibility metrics [25], and Hao et al. proposed a similar structure integrating EV scheduling with distribution network loss minimization [26]. In parallel, equipment-level research has shown that structural design at the device layer can significantly impact system-level energy performance. For example, Zhao et al. [27] investigated the optimal electrode configuration for a compactly assembled industrial alkaline water electrolyzer, whose result highlights the importance of coupling hardware optimization with system-level scheduling and control, particularly in renewable-powered hydrogen production scenarios. These studies illustrate the benefits of decoupled modeling, though they rarely address coordination with equipment maintenance or dynamic reconfigurability.

This study addresses the following research questions:

(1) The first question concerns the co-optimization of grid reconfiguration and source–load dispatch under renewable energy uncertainty.
(2) The second question focuses on how to evaluate and incorporate flexibility into dispatch decisions to improve robustness and efficiency.

To answer these, a two-layer optimization model is proposed. The upper layer optimizes network topology based on branch flexibility, while the lower layer performs distributed robust dispatch considering uncertainty. A hybrid metaheuristic algorithm (ACO–FHO–DE) is developed for efficient solution. Compared with existing approaches, our model integrates reconfiguration, flexibility assessment, robust scheduling, and maintainability in a unified framework, validated on the CPS62 system with Monte Carlo scenarios.

The remainder of this paper is structured as follows: Section 2 describes the modeling framework and algorithm design. Section 3 presents simulation setups and comparative results. Section 4 discusses practical implications and deployment scenarios. Section 5 concludes the study with future work suggestions.

2. System Model and Two-Layer Optimization Framework for Grid-Based Active Distribution Networks

2.1. Upper-Layer Model

The upper-layer model optimizes with the branch flexibility index and network loss as objectives. If the two are directly and simply added, it will affect the optimization effect. As shown in Equation (1):

$$\max\left\{\frac{C^{loss}-min(C^{loss})}{max(C^{loss})-min(C^{loss})}-\frac{F^{BF}-min(F^{BF})}{max(F^{BF})-min(F^{BF})}\right\} \tag{1}$$

$$\begin{cases} F^{BF}=\sum\limits_{\forall c\in N_c} F_c^{BF} \\ C^{loss}=\sum\limits_{t=1}^{T} c^{loss}P_t^{loss} \end{cases}, \tag{2}$$

where F^{BF} is the branch flexibility adequacy of the distribution network, C^{loss} is the network loss cost of the distribution network, P_t^{loss} is the power loss on the distribution network line at time t, and c^{loss} is the network loss cost.

The relevant constraint models of the upper-layer model are as follows:

(1) The traditional Disflow power flow model is a simplified power flow calculation method widely used in distribution network analysis, proposed by Baran and Wu [28]. It simplifies the three-phase power flow equations into a single-phase model based on the characteristics of radial distribution networks, converting nonlinear power flow relationships into linear or quasi-linear equations. This simplification significantly reduces computational complexity, making it suitable for distribution network reconfiguration and optimization problems where rapid solution iteration is required. However, the original Disflow model assumes a fixed network topology, which is not adaptable to the continuous adjustments in distribution network reconfiguration. The traditional Disflow power flow constraints are not adaptable to the continuous adjustments in distribution network reconfiguration, so the model is optimized. By introducing the line-switching 0–1 variable α_l, the equations are relaxed to form equations that are more suitable for the operational changes in the distribution network:

$$\begin{cases} \sum\limits_{l(j,:)\in\Omega^b} \alpha_l P_{l,t}-\sum\limits_{l(:,j)\in\Omega^b} \alpha_l (P_{l,t}-r_l L_{l,t})=P_{j,t} \\ \sum\limits_{l(j,:)\in\Omega^b} \alpha_l Q_{l,t}-\sum\limits_{l(:,j)\in\Omega^b} \alpha_l (Q_{l,t}-x_l L_{l,t})=Q_{j,t} \end{cases}, \tag{3}$$

$$V_{j,t}=V_{i,t}-2(P_{l,t}r_l+Q_{l,t}x_l)+(r_l^2+x_l^2)L_{l,t}\forall l(i,j)\in\Omega^b, \alpha_l=1, \tag{4}$$

$$\left\|\begin{matrix} 2P_{l,t} \\ 2Q_{l,t} \\ L_{l,t}-V_{j,t} \end{matrix}\right\|_2 \le L_{l,t}+V_{j,t} \tag{5}$$

where α_l is a 0–1 variable representing the switching status of branch l, in which α_l = 1 indicates the branch l is in the closed state. $P_{l,t}$, $Q_{l,t}$ are the net injected active power and reactive power at node I at time t, respectively. $P_{j,t}$, $Q_{j,t}$ are the net injected active power and reactive power at the node in period. The resistance and reactance of branch l are denoted as r_l, x_l. $L_{i,t}$ is the square of the current flowing through branch l at time t. $V_{i,t}$ is the square of the voltage at node I at time t; Ω^b represents the set of distribution network branches.

Since the above model is a non-convex model, convex relaxation is performed on the above power flow model, and the final equation is as follows:

$$\begin{cases} V_{j,t} \leq M_4(1-\alpha_l) + V_{i,t} - 2(P_{l,t}r_l + Q_{l,t}x_l) + ({r_l}^2 + {x_l}^2)L_{l,t} \\ V_{j,t} \geq -M_4(1-\alpha_l) + V_{i,t} - 2(P_{l,t}r_l + Q_{l,t}x_l) + ({r_l}^2 + {x_l}^2)L_{l,t} \end{cases}, \tag{6}$$

where M_4 is a sufficiently large positive number.

(2) System security constraints:

$$0 \leq L_{l,t} \leq (I_l^{max})^2, \tag{7}$$

$$-P_l^{max} \leq P_{l,t} \leq P_l^{max}, \tag{8}$$

$$(U_i^{min})^2 \leq V_{i,t} \leq (U_i^{max})^2, \tag{9}$$

where I_l^{max}, I_l^{min} represent the upper and lower limits of the current in branch l, respectively. U_i^{max}, U_i^{min} represent the upper and lower limits of the voltage at node i, respectively. Finally, P_i^{max} represents the active power upper limit of branch l.

(3) Power constraints for injection into the substation:

$$\begin{cases} P_i^{sub,min} \leq P_{i,t}^{sub} \leq P_i^{sub,max} \\ Q_i^{sub,min} \leq Q_{i,t}^{sub} \leq Q_i^{sub,max} \end{cases}, \tag{10}$$

where at node i, $P_i^{sub,max}$, and $P_i^{sub,main}$ represent the maximum and minimum values of active power during transmission, respectively. $Q_i^{sub,\quad max}$ and $Q_i^{sub,min}$ represent the maximum and minimum values of reactive power during transmission, respectively.

2.2. Lower-Layer Model

The lower-layer model uses the grid topology from the upper layer. It considers the uncertainty of wind and solar output to find the worst-case operation strategy. This is achieved by analyzing historical data of loads and renewable energy. The distributionally robust model involved in this paper is a two-stage optimization problem, expressed as follows:

$$\min_x \left(f_1(x) + \max_{p_k \in \Omega} \sum_{k=1}^{K} p_k \min_{y_k \in U(x,\xi_k)} f_2(y_k) \right), \tag{11}$$

$$f_1(x) = C_{sub} + C_{GT} + C_{ESS}, \tag{12}$$

$$\begin{cases} C_{sub} = \sum\limits_{t=1}^{T} \sum\limits_{i \in N^{sub}} c_t^{sub} P_{i,t}^{sub} \\ C_{GT} = \sum\limits_{t=1}^{T} \sum\limits_{i \in N^{GT}} c^{GT} P_{i,t}^{GT} \\ C_{ESS} = \sum\limits_{t=1}^{T} \sum\limits_{i \in N^{ESS}} c^{ESS} (P_{i,t}^{ch} + P_{i,t}^{dis}) \end{cases}, \tag{13}$$

$$f_2(y) = C_{TL} + C_{pv} + C_{WT} + C_{flex}, \tag{14}$$

$$\begin{cases} C_{TL} = \sum_{t=1}^{T} \sum_{i \in N^{TL}} c^{TL} P_{i,t}^{TL} \\ C_{pw} = \sum_{t=1}^{T} \sum_{i \in N^{pw}} c^{pw} (\hat{P}_{i,t}^{pw} - P_{i,t}^{pw}) \\ C_{pv} = \sum_{t=1}^{T} \sum_{i \in N^{pv}} c^{pv} (\hat{P}_{i,t}^{pv} - P_{i,t}^{pv}) \\ C_{flex} = \sum_{t=1}^{T} c^{flex} (F_t^{flex,up} + F_t^{flex,dn}) \end{cases}, \tag{15}$$

where x and y are first-stage and second-stage variables. K is the number of discrete scenarios after sample reduction. ξ_k, p_k are the k-th scenario and its occurrence probability. Ω is the comprehensive norm ambiguity set. C_{sub}, C_{GT}, C_{ESS}, C_{TL}, C_{pw}, C_{pv}, and C_{flew} are the main grid power purchase cost, gas turbine (GT) operation cost, energy storage system (ESS) operation cost, transferable load response cost, wind curtailment cost, solar curtailment cost, and flexibility deficit loss cost, respectively. Finally, ${c_t}^{sub}$, c^{GT}, c^{ESS}, c^{TL}, c^{vw}, c^{pv}, and c^{flex} are the unit prices of power purchased from the upper-level grid, gas turbine generation, ESS operation and maintenance, transferable load response, wind curtailment, solar curtailment, and flexibility deficit loss cost coefficient at time t, respectively.

The model uses historical data and clustering to select K typical scenarios from M samples. These scenarios represent possible uncertainty conditions and construct an initial provability distribution of the uncertainty set. However, due to the limitations of historical data, the initial probability distribution obtained deviates from the actual situation. Therefore, a comprehensive norm constraint is further introduced to construct a set as shown in Equation (16):

$$\Omega = \left\{ \{p_k\} \left| \begin{array}{l} p_k \geq 0, k = 1, \ldots, K \\ \sum_{k=1}^{K} p_k = 1 \\ \sum_{s=1}^{K} \left| p_k - p_k^0 \right| \leq \theta_1 \\ \max_{1 \leq k \leq K} \left| p_k - p_k^0 \right| \leq \theta_\infty \end{array} \right. \right\}, \tag{16}$$

where θ_1 and θ_∞ are the probability allowable deviation limits under 1-norm and ∞-norm constraints, respectively.

The confidence constraint can be constructed based on the given confidence level and historical data volume as shown in Equation (17):

$$\begin{cases} Pr\left\{ \sum_{k=1}^{K} \left| p_k - p_k^0 \right| \leq \theta_1 \right\} \geq 1 - 2Ke^{-2M\theta_1/K} \\ Pr\left\{ \max_{1 \leq k \leq K} \left| p_k - p_k^0 \right| \leq \theta_\infty \right\} \geq 1 - 2Ke^{-2M\theta_\infty} \end{cases}. \tag{17}$$

Let the right-hand sides of the above inequalities be equal to α_1 and α_∞, respectively, as shown in Equation (18):

$$\begin{cases} \theta_1 = \frac{K}{2M} In \frac{2K}{1-\alpha_1} \\ \theta_\infty = \frac{1}{2M} In \frac{2K}{1-\alpha_\infty} \end{cases}, \tag{18}$$

where values θ_1 and θ_∞ represent the maximum degree of deviation between the distributionally robust optimization (DRO) scenario probabilities and the initial scenario probabilities.

An increase in these values implies that the robust model tends to be more conservative, while a decrease assumes more risks. The confidence set of probability distributions in this paper is subject to the joint constraints of 1-norm and ∞-norm, which effectively avoids the occurrence of extreme cases.

In this paper, robustness refers to the ability of the optimization model to maintain feasible and near-optimal solutions under renewable generation and load uncertainty. The proposed model adopts a DRO framework that does not rely on exact probability distributions, but instead uses an ambiguity set to cover a range of possible scenario deviations. This formulation ensures that the dispatch plan remains effective even in worst-case or highly uncertain scenarios.

2.3. Two-Layer Optimization Model

The two-layer optimization model established in this paper is shown in Figure 1.

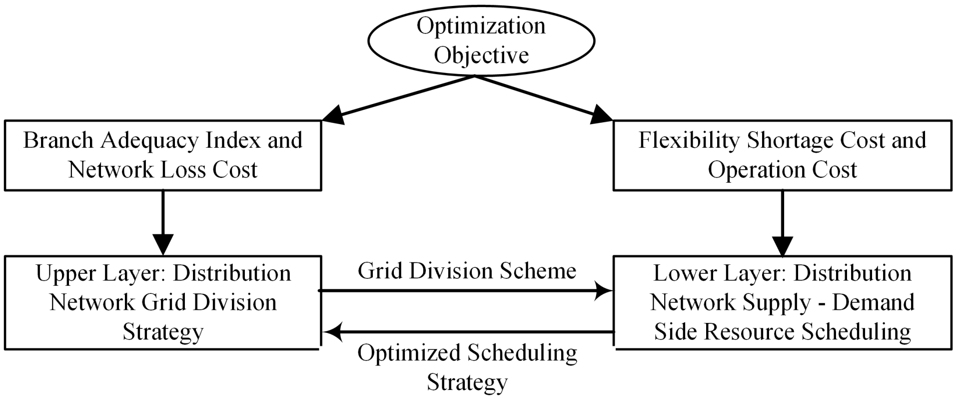


Figure 1. Double-layer optimization structure of grid distribution network.

The two-layer model uses a centralized distributed optimization structure, where the upper layer uses centralized optimization and the lower layer uses distributed optimization for each grid. This structure meets the operational needs of grid-based distribution networks. The top-level design of the model is based on branch flexibility indicators and network loss costs; it applies strategies to determine the optimal division method of the distribution network structure. The bottom-layer model focuses on dispatching adjustable resources on both the supply and demand sides to reduce the risk of system flexibility shortages. It feeds the dispatch results back to the top-layer model. Based on the feedback, the top layer adjusts its strategy and determines a new optimal grid division scheme. Through the mutual iteration of the upper- and lower-layer models, the solution to the problem is achieved.

The upper- and lower-layer models contain a large number of variables. Therefore, a two-layer iterative hybrid optimization algorithm is used. The algorithm combines ACO, FHO, and DE. It also introduces Tent chaos mapping for population initialization and an adaptive weight mechanism. This hybrid algorithm fully utilizes the global search capability of ACO, the local optimization capability of FHO, and the global exploration capability of DE [29–31]. Tent chaotic mapping can enhance population diversity, and the adaptive weight mechanism balances global and local searches [32,33]. These techniques help the algorithm handle multivariable optimization problems in complex models while also ensuring that high-quality solutions are found within a reasonable computational time. As shown in Figure 2, the optimization process follows a two-layer iterative architecture. The algorithm begins with grid initialization and upper-layer topology optimization, targeting the minimization of power loss and the maximization of the flexible branch factor. The resulting topology is then passed to the lower layer, where multi-scenario robust dispatch is performed to minimize operational cost, renewable curtailment, and flexibility degradation. The hybrid optimization algorithm employs a global-local strategy: ACO is used to explore the initial solution space, FHO performs local refinement, and convergence is iteratively checked. If convergence is not achieved, the updated solution is fed back to the upper layer for re-optimization. This feedback loop continues until the system reaches convergence. The final output yields the co-optimized topology and dispatch scheme that balances physical structure and operational efficiency under uncertainty.

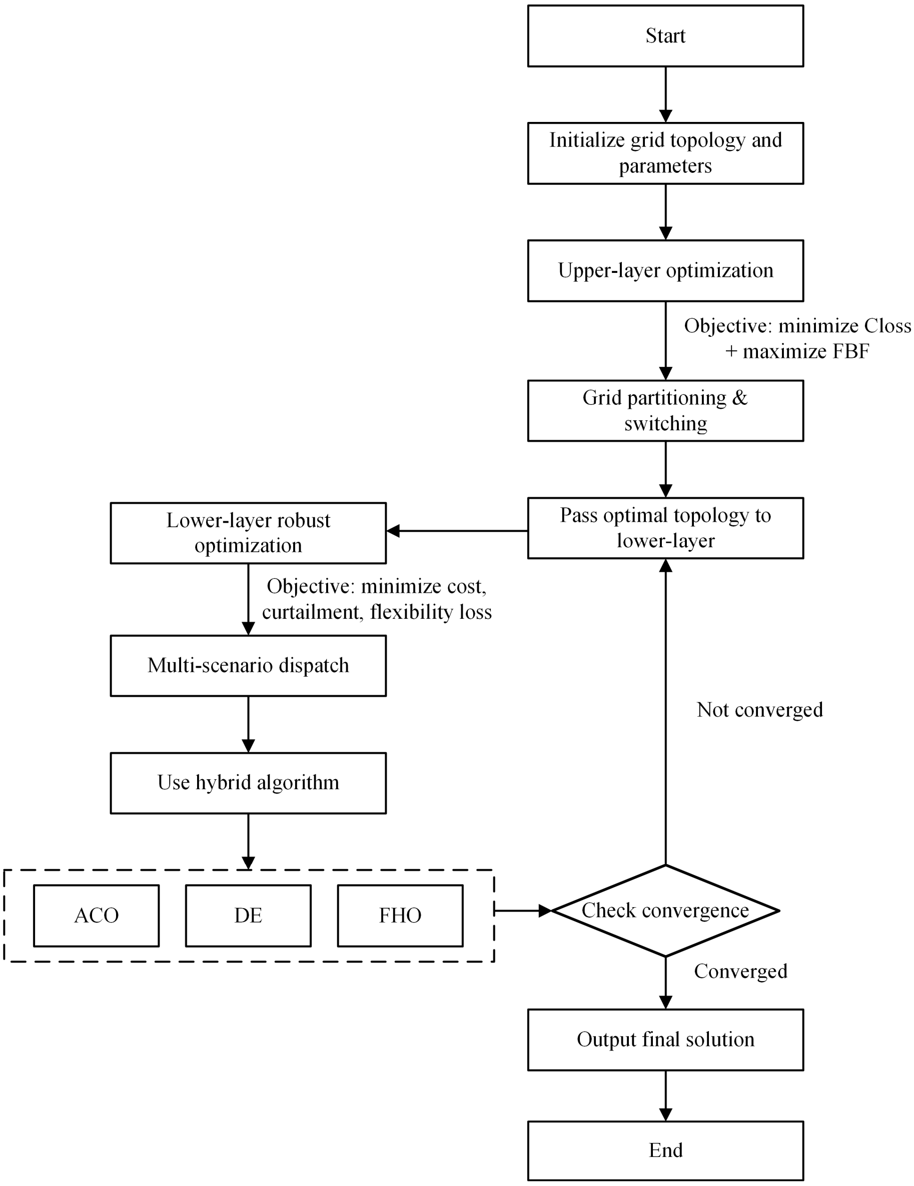


Figure 2. Double-layer iterative solution process.

(1) The ant colony algorithm generates the initial solution.

The Tent chaos mapping is utilized to generate the initial population X_i^0:

$$x_{n+1} = \begin{cases} \frac{x_n}{\mu}, 0 \le x_n < \mu \\ \frac{1-x_n}{1-\mu}, \mu \le x_n \le 1 \end{cases}, \tag{19}$$

$$X_i^0 = X_{min} + x_i \cdot (X_{max} - X_{min}), \tag{20}$$

where x_n represents the n-th value of the chaotic sequence. μ is the control parameter, usually taken as $\mu = 0.5$. x^{min} and x^{max} are the lower and upper bounds of the solution space, respectively.

The path selection probability $P_{ij}^k(t)$ of the ant colony algorithm is used to construct paths and generate the initial solution S^k. The objective function value $f(S^k)$, such as the flexibility index of the distribution network or the network loss cost, is then calculated:

$$P_{ij}^k(t) = \frac{[\tau_{ij}(t)]^\alpha \cdot [\eta_{ij}]^\beta}{\sum\limits_{l \in \mathcal{N}_i^k} [\tau_{il}(t)]^\alpha \cdot [\eta_{il}]^\beta}, \tag{21}$$

where $P_{ij}^k(t)$ denotes the probability that the k-th ant selects node j from node i at time t. $\tau_{ij}(t)$ is the pheromone concentration from node i to node j. η_{ij} is the reciprocal of the path length α and β represent the importance parameters of pheromone and heuristic information, respectively. N_i^k is the set of optional nodes for ant k at node i.

(2) Local optimization by FHO algorithm

The initial solution S^k generated by the ACO is used as the initial positions X_i^0 of the fire hawks and prey. Perform local optimization using the adaptive weight and the position update formula of the FHO algorithm, X_i^t is the position of the i-th fire hawk or prey at time t.

$$X_i^{t+1} = w(t) \cdot X_i^t + \Delta X_i^t \tag{22}$$

Then, calculate the updated objective function value $f(X_i^{t+1})$.

(3) Differential Evolution Strategy

Perform mutation, crossover, and selection operations of differential evolution on the positions updated by the FHO algorithm to generate new candidate solutions. Calculate the objective function value $f(U_i^t)$ of the new candidate solutions:

$$V_i^t = X_{r1}^t + F \cdot (X_{r2}^t - X_{r3}^t), \tag{23}$$

$$U_{ij}^t = \begin{cases} V_{ij}^t, \ \mathit{if\ rand} \le CR \ \mathit{or}\ j = j_{rand} \\ X_{ij}^t, \ \mathit{otherwise} \end{cases}, \tag{24}$$

where V_i^t denotes the mutation vector of the i-th individual. F is the scaling factor, typically taking values in [0, 2]. U_{ij}^t, V_{ij}^t and X_{ij}^t are the j-th dimension of the trial vector, mutation vector, and current individual, respectively. Finally, CR represents the crossover probability, usually within [0, 1].

(4) Combination of Pheromone and Position Updates

Based on the positions updated by the DE strategy, the pheromone enhancement amount is calculated by

$$\Delta\tau_{ij}(t) = \frac{Q}{f(U_i^t)}, \tag{25}$$

$$\tau_{ij}(t+1) = (1-\rho) \cdot \tau_{ij}(t) + \Delta\tau_{ij}(t), \tag{26}$$

where Q is the pheromone enhancement constant.

(5) Iterative Optimization

Repeat steps (1) to (4) until the convergence condition is met.

3. Case Study and Verification of Distribution System Model

The CPS62-node distribution system used in this study is a benchmark case derived from a real-world medium-voltage grid in China. It features distributed PV, wind, energy storage, and demand-side flexibility resources, along with multiple tie-lines and a radial configuration. These characteristics make it more representative of practical active distribution networks than classical IEEE test systems, which often lack topological complexity and renewable uncertainty modeling [34,35]. The CPS62 system offers a balance between computational tractability and modeling richness, enabling effective evaluation of coordinated reconfiguration and robust dispatch strategies [36].

3.1. Analysis of Optimization Results

This paper designs three different schemes to verify the economy and flexibility of the model. Scheme 1 does not consider the flexibility index or grid interconnection. Scheme 2 includes the flexibility index but does not include grid interconnection. Scheme 3 considers both grid interconnection and the operational flexibility index. Scheme 1 uses the traditional optimization method to optimize only the operation cost. As shown in Table 1, the flexibility deficits of Grid 1 and Grid 2 are relatively high, accompanied by severe wind/solar curtailment and line overload conditions. Table 2 indicates that although the operation cost of Scheme 1 is the lowest among the three schemes, its comprehensive cost is the highest due to the high flexibility deficit cost. Scheme 1 does not include the operational flexibility index or flexible interconnection between grids in the optimization process. As a result, it cannot solve the problem of serious local flexibility shortages caused by the high penetration of new energy sources.

Table 1. Grid indicator results of each scheme.

Scheme	Grid	Curtailed Wind-Solar Energy/(kWh)	Grid Flexibility Deficit	Grid Branch Flexibility Adequacy
One	①	253	0.18	0.72
	②	61	0.31	0.35
	③	0	0	0.86
Two	①	297	0.14	0.67
	②	145	0.17	0.31
	③	0	0	0.86
Three	①	82	0.04	0.62
	②	54	0.06	0.57
	③	9	0.01	0.71

Table 2. Optimization results under different schemes/CNY 10,000.

Parameters and Units		Scheme 1	Scheme 2	Scheme 3
Over Source—side Resource Cost	Main Grid Power Purchase	2.2371	2.2725	2.2751
	Total Network Loss	0.0746	0.0738	0.0654
	MTG Power Generation	0.9163	0.9377	0.9787
	Energy Storage	0.2422	0.2226	0.2137
	Wind—Solar Curtailment	0.0316	0.0444	0.0147
Load—side Resource Cost	Transferable Load Response	0.0950	0.0933	0.0938
Total Operation Cost		3.6346	3.6794	3.6479
Flexibility Deficit Cost		0.1685	0.0815	0.0184
Comprehensive Cost		3.7743	3.7249	3.6686

The simulation test adopts a modified CPS62-node active distribution system, as illustrated in Figure 3. Topology of the modified CPS62-node distribution system with renewable and flexible resources. The system consists of three interconnected grids (Grid 1, Grid 2, and Grid 3) linked by tie-lines (red dashed lines). Each black solid line represents a feeder section, with numbered nodes indicating bus locations. Photovoltaic generation units (blue diamonds) are installed at nodes 14,35 and 49, wind turbines (red triangles) at nodes 6, 26, and 55, and energy storage systems (green squares) at nodes 8, 31, and 50. Load points are located at all numbered buses except those hosting only generation or storage. The unit costs of various source-load resources are shown in Figure 4. The standard value of voltage is set to 12.66 kV, the power reference value is 1 MVA, and the upper limit of transmission capacity for each branch circuit is 1 MW. The active power output of the GT units is capped at 0.3 MW, the ramp rate is limited to 0.1 MW, and the maximum permissible energy capacity of the ESS is 0.9 MWh. The upper limit of charging and discharging power is 0.3 MW. The charging and discharging efficiencies are 0.95. Figure 5 shows the forecasted power profiles, including Figure 5a the predicted output of PV systems and wind turbines, and Figure 5b the load curves for Grid 1, Grid 2, and Grid 3. It should be noted that the wind power profile shown in Figure 5a is a smoothed representative curve used for visualization purposes. In practice, wind power output is inherently intermittent and volatile. This variability is fully accounted for in the model through the generation of 500 uncertainty scenarios using Latin Hypercube Sampling based on historical wind data distributions. These scenarios capture the full range of wind generation fluctuations and are used in the lower-layer robust dispatch to ensure that uncertainty is properly reflected in the optimization process.

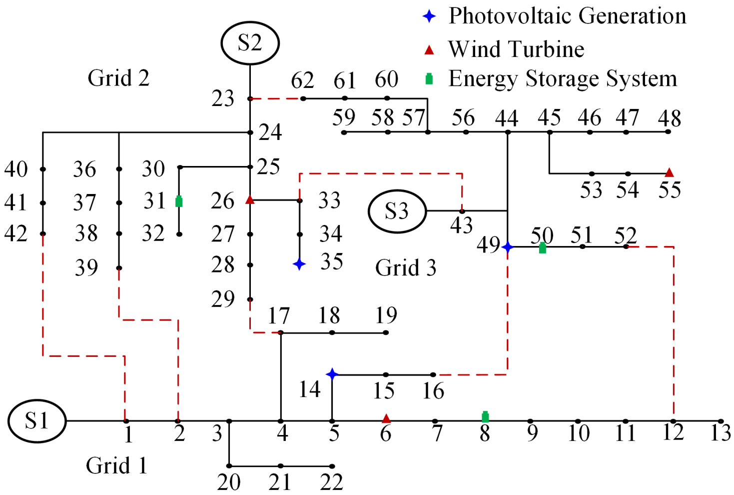


Figure 3. CPS62 node simulation example. The red dotted lines represent normally open tie-lines.

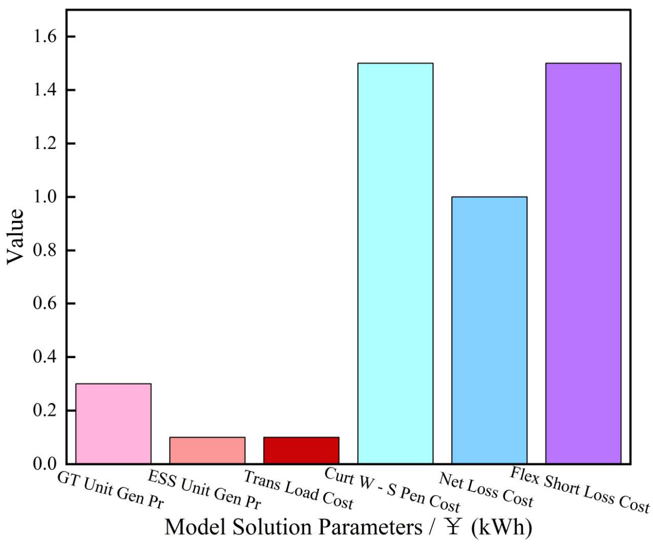
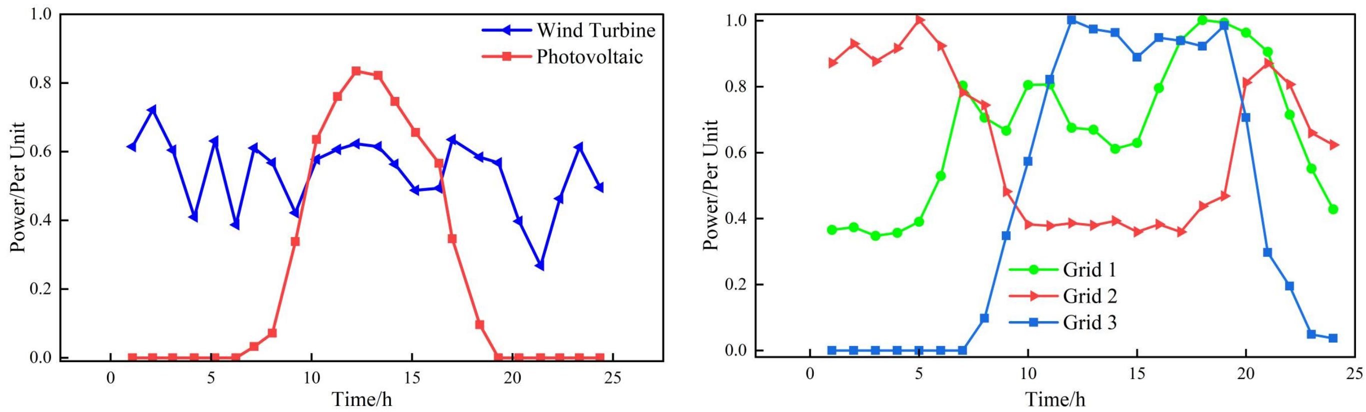


Figure 4. Unit cost of source load resources.



(**a**) (**b**)

Figure 5. Forecasted source and load profiles. (**a**) Output curves of wind and PV. (**b**) Load curves of Grid 1, Grid 2, and Grid 3.

Scheme 2 adopts the initial grid state and takes the flexibility index as the optimization objective. Table 1 shows that the flexibility deficits of Grid 1 and Grid 2 are lower than those in Scheme 1. However, this improvement comes at the cost of reduced wind-solar utilization and decreased branch capacity margin. As a result, the actual problem remains unsolved. The indicators of Grid 3 are almost the same as those in Scheme 1, suggesting that Grid 3 does not fully utilize its own advantages and highlighting an imbalance in dynamic resource allocation among the grids.

Scheme 3 employs the double-layer optimization model proposed in this paper, allowing power interconnection between grids. Figure 6 shows the switching states of lines after grid interconnection. Compared with Scheme 1 and Scheme 2, Scheme 3 achieves a dynamic balance of supply and demand in each grid. It uses flexibility indicators and power interconnection between grids to guide the adjustment process. This approach effectively solves local flexibility shortages that may appear in real power system operations.

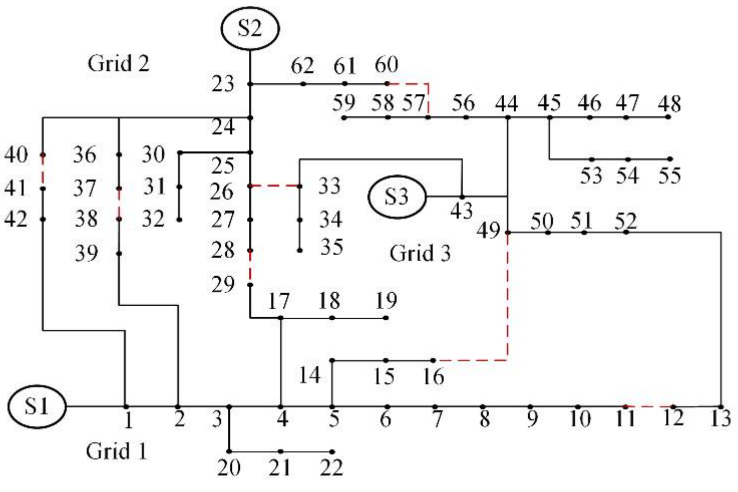


Figure 6. Calculation example after grid interconnection. The red dotted lines represent normally open tie-lines.

Simulation also reveals that renewable energy locations and capacities significantly influence optimization outcomes. For instance, if PVs are unevenly distributed among grids, local flexibility mismatch worsens without proper grid reconfiguration. The model effectively reallocates power paths and adjusts flexibility reserves to mitigate this imbalance. In the proposed framework, the upper-layer optimization determines the most effective grid partitioning and branch switching plan to improve network-wide flexibility. The lower-layer optimization then performs dispatch decisions for each grid under uncertainty. Results show that this interaction reduces curtailment, enhances load balancing, and minimizes risk across all operating scenarios.

A comparative analysis of the three schemes highlights their distinct characteristics and advantages. Scheme 1, which only considers operation costs, demonstrates lower direct cost but suffers from high flexibility deficits and resource curtailment. This shows that it is economical but not suitable under high renewable penetration. Scheme 2 introduces a flexibility index into the optimization but lacks grid interconnection. It reduces flexibility losses but cannot fully leverage flexible resources across the system, resulting in unbalanced performance among grids. Scheme 3, which integrates both flexibility index and grid interconnection based on the proposed two-layer model, achieves the best overall performance. It balances supply and demand dynamically, minimizes curtailment, and reduces total cost, demonstrating superior adaptability under uncertainty. These results confirm that incorporating flexibility indicators and network reconfiguration simultaneously is essential for enhancing economic and operational performance in modern distribution systems.

3.2. Analysis and Comparative Analysis of Robust Optimization Results

In this study, uncertainties primarily arise from renewable generation variability (wind and PV output), load demand fluctuations, and forecasting errors. Disturbances may also include equipment outages, unexpected changes in network topology, and communication delays. The proposed DRO framework accounts for these factors by constructing an ambiguity set that captures deviations from the nominal probability distribution.

To comprehensively evaluate the effectiveness of the proposed optimization method under uncertainty, this study compares four dispatch strategies with distinct modeling philosophies. The deterministic optimization model is based on forecast point values without considering uncertainty, and although simplistic, it provides a baseline reference to quantify the value of uncertainty modeling. The stochastic programming model incorporates multiple probabilistic scenarios with assigned occurrence probabilities, enabling the assessment of expected performance but potentially leading to suboptimal results in rare but severe conditions. In contrast, the traditional robust optimization model protects against the worst-case scenario, ensuring operational feasibility under extreme conditions, though often at the cost of overly conservative solutions and higher average costs. The proposed DRO model adopts an ambiguity set approach to capture uncertainty without requiring precise probability distributions. This allows for a more balanced solution that maintains robustness while improving economic efficiency. The comparative analysis that follows demonstrates the relative strengths and trade-offs of these methods across a wide range of simulated scenarios.

To evaluate the robustness and adaptability of the model under renewable energy uncertainty, we adopt a Monte Carlo simulation approach. A total of 500 synthetic wind–solar–load scenarios are constructed based on historical data patterns, incorporating fluctuations in PV and wind output as well as grid load variations. The scenarios are generated using Latin Hypercube Sampling and then clustered to ensure coverage of a wide range of operating conditions. Among these scenarios, three typical cases are selected for detailed analysis. The first is a high wind–solar generation combined with peak load, which poses a severe challenge to dispatch flexibility due to simultaneous supply and demand surges. The second represents a medium wind–solar output with a flat load profile, corresponding to typical operating conditions. The third scenario involves low renewable energy generation during valley load hours, leading to an increased risk of flexibility deficits and supply inadequacy. These representative cases enable a comprehensive evaluation of the model's performance under various uncertainty conditions. As illustrated in Figure 8, the deterministic model exhibits the highest average and maximum costs due to its inability to account for uncertainty. Although the stochastic optimization model achieves a lower average cost, it lacks robustness, resulting in high operational risks under unfavorable renewable generation and load conditions. The robust model offers better protection against cost spikes, but its overly conservative nature leads to higher comprehensive costs. In contrast, the proposed model achieves the lowest average cost and the lowest maximum cost across all scenarios. These results demonstrate that the proposed approach effectively balances cost and risk, offering superior robustness and economic performance compared with the other three methods.

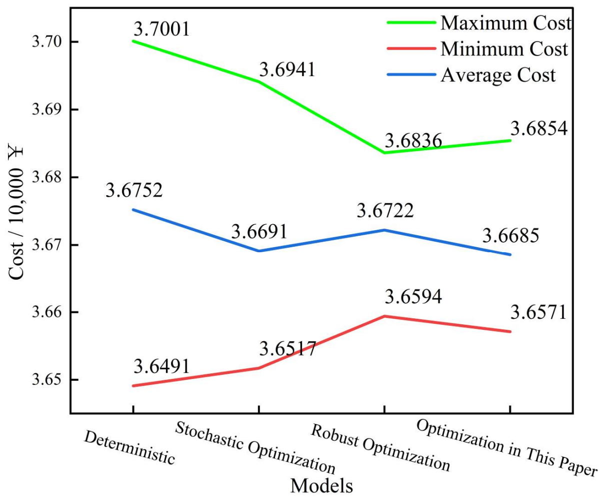


Figure 7. Comparison of Monte Carlo statistical results/CNY 10,000.

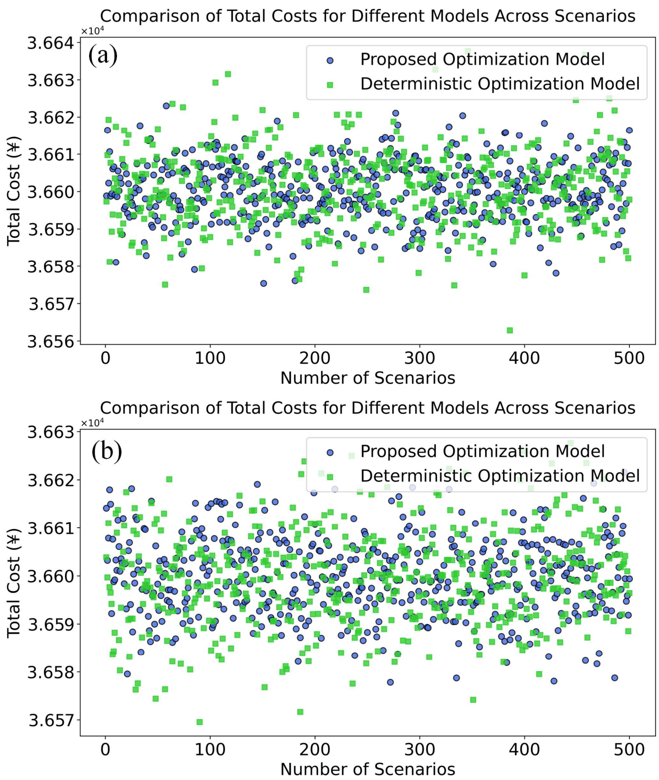


Figure 8. *Cont.*

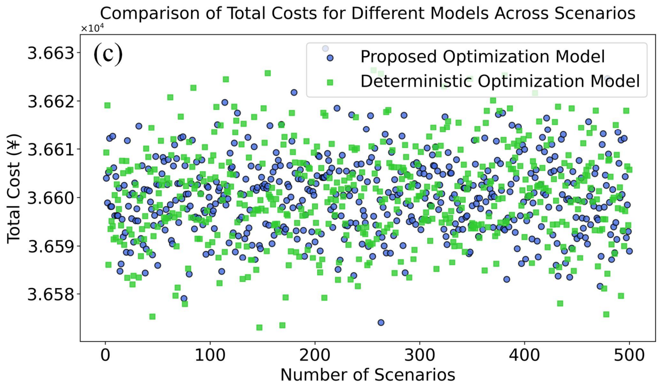


Figure 8. Cost comparison among different optimization models under typical scenarios. (**a**) High wind–solar generation and peak load. (**b**) Medium wind–solar generation and flat load. (**c**) Low wind–solar generation and valley load.

The proposed two-layer optimization model and hybrid algorithm are designed in a modular and scalable way. Therefore, they can be extended to larger systems or systems with different configurations with minor adjustments to parameters and constraints. All modeling, simulation, and optimization were implemented in MATLAB R2021a. The mathematical formulation of the optimization model was encoded using YALMIP (version 0.9.7), while the hybrid algorithm was developed using customized scripts. All simulations were run on a standard desktop computer with Intel i7 CPU and 32 GB RAM.

To evaluate the robustness of the proposed model, we compare the distribution of total operating cost across 500 scenarios for four models: deterministic, stochastic, traditional robust, and proposed DRO. Two performance indicators are used: Maximum cost across all scenarios (reflects worst-case behavior). Average cost across all scenarios (reflects overall performance stability). As shown in Figure 8, the deterministic model has the highest maximum and average cost due to its lack of uncertainty handling. The stochastic model performs well on average but exhibits cost spikes under adverse conditions. The traditional robust model is safer but overly conservative. The proposed model achieves the lowest maximum cost (i.e., best worst-case performance) and a moderate average cost, confirming its superior robustness and cost-effectiveness.

Several existing studies have addressed the problem of scheduling in uncertain distribution networks. For instance, Yi et al. [17] proposed a multi-objective robust optimization model incorporating demand response uncertainty, achieving 4.2% cost reduction in IEEE 33-bus system under 200 scenarios. Liao et al. [18] employed affine adjustable robust control and showed improved feasibility under deep renewable penetration. Huang et al. [19] developed a web-of-cells-based two-layer optimization, achieving balanced flexibility allocation but with higher average cost due to lack of loss minimization.

Compared with these studies, the proposed method achieves both lower average and worst-case cost across 500 scenarios, and additionally considers equipment maintainability through branch flexibility evaluation. As shown in Table 2 and Figure 8, our model achieves an average cost reduction of 6.8% compared with traditional robust methods, and up to 14.5% compared with deterministic optimization. These results confirm the model's superior adaptability and economic performance under uncertainty.

4. Discussion

The simulation results confirm that the proposed two-layer optimization model effectively enhances both economic performance and operational flexibility under renewable energy uncertainty. Compared with deterministic, stochastic, and classical robust optimization models, the proposed approach consistently achieves lower average and maximum operating costs while reducing renewable energy curtailment, flexibility deficits, and system risk. These improvements are mainly attributed to the coordinated design of flexibility evaluation and grid reconfiguration in the upper layer, coupled with robust dispatch in the lower layer. This synergy enables adaptive adjustment to fluctuating operating conditions without excessive conservatism. The proposed method is theoretically grounded in bilevel optimization theory, which decomposes structural and operational decision-making to improve computational tractability. The upper-layer reconfiguration problem leverages graph theory to represent network topology and employs a flexibility adequacy index to quantify the network's capacity to accommodate variability and disturbances. The lower-layer dispatch adopts a distributionally robust optimization formulation, ensuring feasible and cost-effective operation under worst-case probability distributions within an ambiguity set. This integration of structural planning and operational scheduling exploits the hierarchical nature of active distribution networks and directly addresses spatial-temporal flexibility allocation.

In practical applications, the model also enhances maintainability by using branch flexibility indices and switchable topology to support predictive and condition-based maintenance through improved load balancing and branch stress monitoring. Its modular design and scenario-based formulation ensure strong scalability and adaptability across various system configurations, including microgrids, rural networks, and large-scale urban systems.

To implement this model in real-world distribution systems, three key infrastructures are required:

(1) Topology reconfiguration capabilities;
(2) Monitoring and forecasting infrastructure;
(3) Controllable flexible resources.

The applicability of the model is based on several assumptions: availability of historical or forecasted renewable and load profiles; known and controllable grid topology and equipment constraints; coordinated control of flexible loads and distributed resources; and sufficient computational resources to solve the dispatch problem offline or in near real-time.

Overall, the proposed model demonstrates significant potential for practical deployment in renewable-rich active distribution networks. It not only improves cost efficiency and operational resilience but also provides a decision-support framework that can be integrated into future energy management systems (EMS) and distribution automation systems (DAS).

5. Conclusions

This paper proposes a two-layer grid-based optimization method for distribution networks with high renewable energy penetration. The upper layer optimizes network topology to enhance branch flexibility and minimize power losses, while the lower layer performs distributed robust dispatch under uncertainty to reduce operational costs. A hybrid algorithm combining Ant Colony Optimization, Fire Hawk Optimization, and Differential Evolution is used to solve the model efficiently. Simulation results on a 62-node system demonstrate that the proposed method reduces total costs, improves flexibility, and effectively mitigates renewable curtailment and equipment stress. These outcomes confirm the model's potential to support smart grid scheduling and predictive maintenance. The approach is particularly suited for Distribution System Operators, offering a practical decision-support tool that coordinates grid reconfiguration and flexibility management. It also benefits distributed energy operators by enhancing resource utilization and reducing curtailment.

Despite the promising results, several limitations remain in the current work. First, the proposed model relies on the availability and accuracy of historical data for renewable generation and load profiles, which may limit reliability in data-sparse regions. Second, the optimization framework assumes centralized coordination and controllability of distributed resources, which may not always be achievable due to infrastructure constraints. Third, while the hybrid metaheuristic algorithm offers effective global and local search capability, it may encounter scalability challenges when applied to ultra-large distribution systems or real-time operation scenarios.

Future research will focus on improving real-time applicability through algorithmic acceleration, integrating demand-side behavioral uncertainty, and extending the uncertainty modeling to cover cyber–physical risks such as communication delays and sensor failures. Moreover, incorporating adaptive coordination strategies under partially observable environments and validating the model on actual distribution networks with supervisory control and data acquisition integration will be key directions for practical deployment.

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Abbreviations

List of abbreviations and expansions.

DRO	Distributionally Robust Optimization
ACO	Ant Colony Optimization
FHO	Fire Hawk Optimization
DE	Differential Evolution
ESS	Energy Storage System
PV	Photovoltaic
GT	Gas Turbine

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